

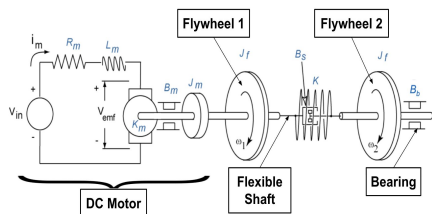
Introduction and Objectives

Mesa Hybrid Vehicles Inc. is experiencing vibration issues with the drive shaft of the *Gulo 3000*, a new hybrid prototype.

We have been asked to:

- Verify drive shaft flexibility as the vibration source
- Characterise the system and develop a math model
- Use a validated model to identify 4 solutions to reduce the vibrations by 30%

Experimental Setup and Procedure

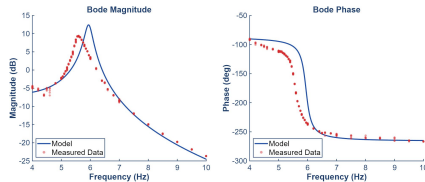


Damping Coefficients: Experiments were performed to determine the damping coefficients of parts of the system.

Simulation Modeling: Used our data to generate a transfer function model to simulate our system and test parameters.

System Model

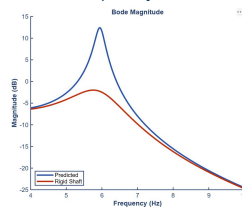
Frequency Response Validation



Shaft Flexibility Causes Vibration

Three factors verified shaft flexibility as the main source of vibrations:

- Sensitivity analysis conducted to determine effect of each model parameter on the transmissibility around the resonant frequency
- Tested each parameter to see if it resulted in a change in vibration magnitude or natural frequency
- Compared prediction against model with zero shaft damping to imitate a rigid shaft

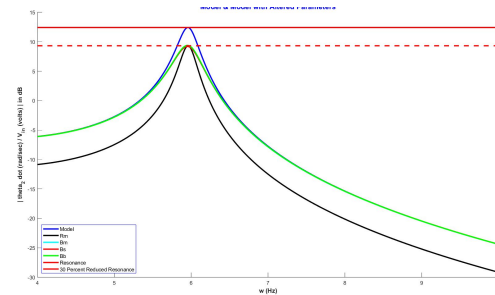


Vibrational Quantification

- Sensitivity Analysis
 - Cost function
 - Magnitude change vs shift
- Provide the process of how tested the different solutions to reduce vibrations
- Give graphs and reduction %

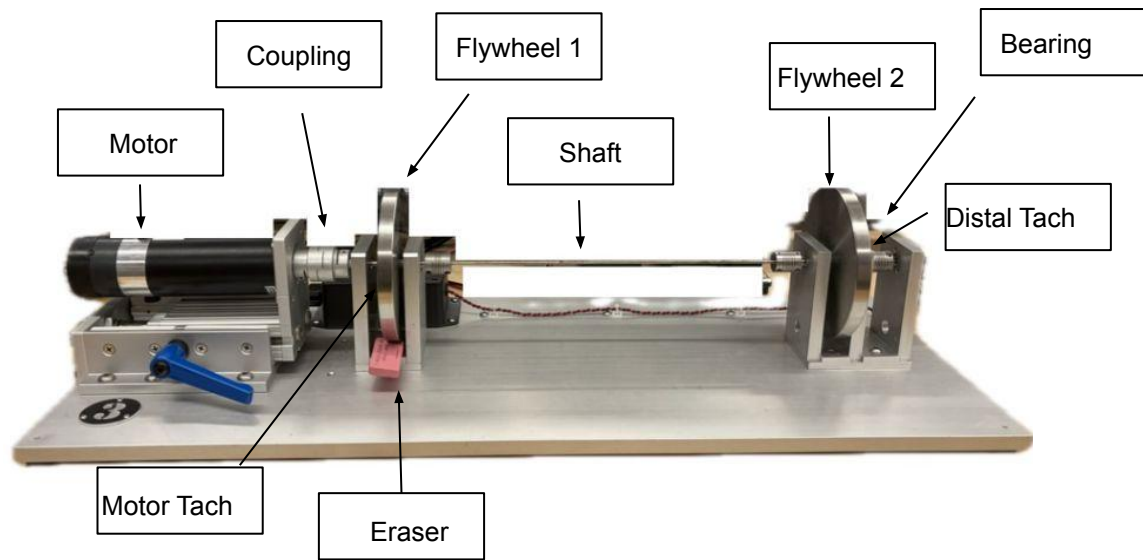
Design Recommendations

1. **Add lubrication to bearing** (Increase motor damping)
2. **Add resistor in series with motor armature** (Increase motor resistance)
3. **Add torsional damper to shaft** (Increase shaft damping)
4. **Add inertia clean body damper to motor** (Increase motor damping)



Citations

- [1] T. Matsuoka, "Electromagnetic damper using brushless direct drive DC motor," *Mech. Eng. J.*, vol. 5, no. 6, p. 18-00307, 2018.
- [2] Wang, Zhicheng, et al. "Experimental Study on the Skyhook Control of a Magnetorheological Torsional Vibration Damper." *Micro Machines* 15.2 (2024): 236.
- [3] Napolitano, K.L., et al. "A Comparison of Two Cured Damped Composite Torsion Shafts." *Composite Structures* 43.2 (1998): 115-125.



https://docs.google.com/document/d/1IV0jsExRMIBDRi4BuH4hX4QXIS_GagDMV_FzAq2m2DFM/edit?usp=sharing

<https://docs.google.com/document/d/1q7ViSLsunoh-CdXCT0qXqIFHsjvDlrHhIDrxZUbUIkE/edit?usp=sharing>

<https://docs.google.com/document/d/1z8yMJc19LrDomBsEDnbWvBpRxJTj7aDiLfOfYesFpPU/edit?usp=sharing>

<https://docs.google.com/document/d/1B2tYq432aKqoXG3oq9hUTLoixHSv9pccfmbgnb0BXhk/edit?tab=t.0#heading=h.exa4gu26d7s9>



Fixes for the poster 1st document

- Figure axis labels are too small
- Header font size needs to be same, headers need to line up
- Provide labels and bigger legends on figures
- Need same amount of horizontal space between headers
- Title doesn't stand out, maybe shorten it, needs to be better, remove "improve", mention "reduce vibrations" before main topic
- Need logo
- Could change some headings (vibrational quantification to vibration measurements? and recommendations to design recommendations?)
- Mentioned the damping coefficients before why we needed them
- Put simulation model before missing damping coefficients
- Figure text bigger
- Need to mention what 3 factors explain draft shaft flexibility as vibration source
- Need to use different colors, reused color red 3x, maybe change using green to another color
- Make recommendation graph bigger, zoom in
- Need to quantify the % of vibration reduction

Fixes for the poster 2nd document

- No Logos
- Make sure text doesn't extend out of section
- Says the title is good but should add "vehicle" instead of "model"
- More space is needed for the description of "vibrational quantification"
- Text for the visuals are hard to read, zoom in on the graphs, but requires us to remove some text and add it to the handout
- The caption under the "system model" is too small, maybe add labels on the actual figure
- Make heading font size the same
- Redo "Simulation Modeling: Used our data to generate a model to simulate our physical system and test parameters." & "Tested each parameter to see if it resulted in a change in vibration magnitude or natural frequency" (how is this a factor)
- We should define the most important parameters
- The "Vibrational quantification section" may be redundant, its purpose is meant for the recommendation section
- Try using the ylim function in matlab to zoom in on the plot and leave out extra white space
- The texts on the figures could be slightly bigger
- Make the figures larger to be readable
- Need more practicable recommendations that we can use to quantify the % of vibrations

Fixes for the poster 3rd document

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Fixes for the poster 4th document

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